Vector Calculus, Linear Algebra and Differential Forms: A Unified Approach - Solutions

June 28, 2026

Contents

0 Preliminaries **1**
- 0.2 Quantifiers and Negation . . . 1
- 0.3 Set Theory . . . 1
- 0.4 Functions . . . 2
- 0.5 Real Numbers . . . 5
- 0.6 Infinite Sets . . . 8

Chapter 0

Preliminaries

0.2 Quantifiers and Negation

Exercise 0.2.1 Negate the following statements:

a. Every prime number such that if you divide it by 4 you have a remainder of 1 is the sum of two squares.

b. For all $x \in \mathbb{R}$ and for all $\epsilon > 0$, there exists $\delta > 0$ such that for all $y \in \mathbb{R}$, if $|y-x| < \delta$, then $|y^2-x^2| < \epsilon$.

c. For all $\epsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in \mathbb{R}$, if $|y-x| < \delta$, then $|y^2-x^2| < \epsilon$.

SOLUTION

a. There exists a prime number in the form of $4n+1$, for $n \in \mathbb{N}_0$ that cannot be expressed as the sum of two squares.

b. There exists $x \in \mathbb{R}$ and $\epsilon > 0$ such that for all $\delta > 0$, there exists $y \in \mathbb{R}$ with $|y-x| < \delta$ but $|y^2-x^2| \geq \epsilon$.

c. There exists $\epsilon > 0$ such that for all $\delta > 0$, there exists $x, y \in \mathbb{R}$ with $|y-x| < \delta$ but $|y^2-x^2| \geq \epsilon$.

Exercise 0.2.2 Explain why one of these statements is true and the other is false:

$$(\forall \text{ man } M)(\exists \text{ woman } W) \mid W \text{ is the mother of M}$$
$$(\exists \text{ woman } W)(\forall \text{ man } M) \mid W \text{ is the mother of M}$$

SOLUTION

The difference between the two statements lies in the quantifier. Obviously the first statement is true while the second is false.

The first statement states that for every man, we can find a woman that is the man's father. This is inherently true as every man must have a mother.

On the other hand, the second statement states that we can find a women that is the mother of every man, which is obviously not possible.

0.3 Set Theory

Exercise 0.3.1 Let E be a set, with subsets $A \subset E$ and $B \subset E$, and let $*$ be the operation $A * B = (E-A) \cap (E-B)$. Express the sets

a. $A \cup B$ b. $A \cap B$ c. $E - A$

using A, B and $*$.

SOLUTION

Before solving this problem, consider the special case of the operator $*$:

$$X * X = (E-X) \cap (E-X) = X^c \cap X^c = X^c$$

a. We can first simpify the operator with $A * B = A^c \cap B^c = (A \cup B)^c$. Using the special case, we get

$$(A * B) * (A * B) = (A \cup B)^c * (A \cup B)^c = A \cup B$$

b. Working backwards, we get

$$A \cap B = (A^c \cup B^c)^c = ((A * A) \cup (B * B))^c = (A * A) * (B * B)$$

c. This case is the equivilant to the special case above, hencec $E - A = A^c = A * A$.

0.4 Functions

Exercise 0.4.1 Are the following true functions? That is, are they both everywhere defined and well defined?

a. "The aunt of", from people to people.

b. $f(x) = \frac{1}{x}$, from real numbers to real numbers.

c. "The capital of", from countries to cities (careful - at least one country, Bolivia, has to capitals.)

SOLUTION

a. Considering the cases where and individual's parents not having any sisters, the individual does not have an aunt, making it not everywhere defined. Considering the cases where an individual has multiple aunts, the function is not well defined.

b. The function is not everywhere defined on real numbers as when $x = 0$, $f(x)$ is undefined. The function however is wel defined as if $f(x) = f(y)$, then $x = y$.

c. The function is everywhere defined as every country has a capital. However, it is not well defined as some countries, such as Bolivia, has two capitals.

Exercise 0.4.2 a. Make up a nonmathematical function that is onto but not one to one.

b. Make up a mathematical function that is onto but not one to one.

SOLUTION

a. The suit of a poker card is onto but not one to one; every card has a suit but more that one card has the same suit.

b. $f : \mathbb{R} \to \mathbb{R} \quad f(x) = x^2$

Exercise 0.4.3 a. Make up a nonmathematical function that is bijective (onto and one to one).

b. Make up a mathematical function that is bijective.

SOLUTION

a. The thumbprint of a person is bijective as everyone has a thumbprint that is unique.

b. $f : \mathbb{R} \to \mathbb{R} \quad f(x) = x$.

Exercise 0.4.4 a. Make up a nonmathematical function that is one to one but not onto.

b. Make up a mathematical function that is one to one but not onto.

SOLUTION

a. The function mapping credit card to credit card numbers is one to one but not onto; as each card number is unique but not all card numbers have been used.

b. $f : \mathbb{R} \to \mathbb{R} \quad f(x) = e^x$.

Exercise 0.4.5 Given the functions $f : A \to B$, $g : B \to C$, $h : A \to C$, and $k : C \to A$, which of the following compositions are well defined? For those that are, give the domain and codomain of each.

a. $f \circ g$	b. $g \circ f$	c. $h \circ f$	d. $g \circ h$	e. $k \circ h$	f. $h \circ g$
g. $h \circ k$	h. $k \circ g$	i. $k \circ f$	j. $f \circ h$	k. $f \circ k$	l. $g \circ k$

SOLUTION

a. Not well defined.	b. $g \circ f : A \to C$	c. Not well defined.	d. Not well defined.
e. $k \circ h : A \to A$	f. Not well defined.	g. $h \circ k : C \to C$	h. $k \circ g : B \to A$
i. Not well defined.	j. Not well defined.	k. $f \circ k : C \to B$	l. Not well defined.

Exercise 0.4.6 Prove Proposition 0.4.11. **Inverse image of intersection, union**

a. The inverse image of an intersection equals the intersection of the inverse images:

$$f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$$

b. The inverse image of a union equals the union of the inverse images:

$$f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$$

SOLUTION

a. We will show this by inclusion both ways:

   i. Let $y \in A \cap B$ with $x = f^{-1}(y) \in f^{-1}(A \cap B)$. Then $y \in A$ and $y \in B$, implying $x \in f^{-1}(A)$ and $x \in f^{-1}(B)$, thus $x \in f^{-1}(A) \cap f^{-1}(B)$.

   ii. The inverse is similar. $x \in f^{-1}(A) \cap f^{-1}(B)$, then $x \in f^{-1}(A)$ and $x \in f^{-1}(B)$, so $y \in A$ and $y \in B$. This shows that $y \in A \cap B$, hence, $x \in f^{-1}(A \cap B)$.

b. Similarly, by inclusion both ways:

   i. Let $y \in A \cup B$ with $x = f^{-1}(y) \in f^{-1}(A \cup B)$. Then $y \in A$ or $y \in B$, implying $x \in f^{-1}(A)$ or $x \in f^{-1}(B)$, thus $x \in f^{-1}(A) \cup f^{-1}(B)$.

   ii. The inverse is similar. $x \in f^{-1}(A) \cup f^{-1}(B)$, then $x \in f^{-1}(A)$ or $x \in f^{-1}(B)$, so $y \in A$ or $y \in B$. This shows that $y \in A \cup B$, hence, $x \in f^{-1}(A \cup B)$.

Exercise 0.4.7 Evaluate $(f \circ g \circ h)(x)$ at $x = a$ for the following.

a. $f(x) = x^2 - 1, \quad g(x) = 3x, \quad h(x) = -x + 2, \quad$ for $a = 3$.

b. $f(x) = x^2, \quad g(x) = x - 3, \quad h(x) = x - 3, \quad$ for $a = 1$

SOLUTION

a.

$$\begin{aligned}(f \circ g \circ h)(3) &= f(g(h(3))) \\ &= f(g(-3+2)) = f(g(-1)) \\ &= f(3(-1)) = f(-3) \\ &= (-3)^2 - 1 = 8\end{aligned}$$

b.

$$\begin{aligned}(f \circ g \circ h)(1) &= f(g(h(1))) \\ &= f(g(1-3)) = f(g(-2)) \\ &= f(-2-3) = f(-5) \\ &= (-5)^2 = 25\end{aligned}$$

Exercise 0.4.8 a. Prove Proposition 0.4.16. **(Composition of onto functions).** Let the functions $f : B \to C$ and $g : A \to B$ be onto. Then the composition $(f \circ g)$ is onto.

b. Prove Proposition 0.4.17. **(Composition of one to one functions).** Let $f : B \to C$ and $g : A \to B$ be one to one. Then the composition $(f \circ g)$ is one to one.

SOLUTION

a. Let $c \in C$. Since f is onto, there exists some $b \in B$ such that $f(b) = c$. Since g is onto, there exists some $a \in A$ such that $g(a) = b$. Hence $(f \circ g)(a) = f(g(a)) = f(b) = c$. Since every $c \in C$ has a preimage under $f \circ g$, the composition $f \circ g$ is onto.

b. Let $a_1, a_2 \in A$ and suppose $(f \circ g)(a_1) = (f \circ g)(a_2)$. Then $f(g(a_1)) = f(g(a_2))$. Since f is one to one, $g(a_1) = g(a_2)$. Since g is one to one, $a_1 = a_2$. Therefore $f \circ g$ is one to one.

Exercise 0.4.9 What is the natural domain of $\sqrt{\dfrac{x}{y}}$?

SOLUTION

For $\sqrt{\frac{x}{y}}$ to be defined, $\frac{x}{y} \geq 0$. Hence the natural domain is $\{(x, y) \in \mathbb{R}^2 : y \neq 0, \frac{x}{y} \geq 0\}$.

Exercise 0.4.10 What is the natural domain of

a. $\ln \circ \ln$? b. $\ln \circ \ln \circ \ln$? c. ln composed with itself n times?

SOLUTION

a. The domain of $\ln x$ is $x > 0$. Hence the domain of $\ln \circ \ln$ is when $\ln x > 0$ which is $\{x \in \mathbb{R} : x > 1\}$.

b. Similarly, the natural domain of $\ln \circ \ln \circ \ln$ is $\{x \in \mathbb{R} : x > e\}$.

c. The natural domain of ln composed with itself n times is $\{x \in \mathbb{R} : x > e \uparrow\uparrow (n - 2)\}$.

Exercise 0.4.11 What subset of $\mathbb{R}$ is the natural domain of the function $(1 + x)^{1/x}$?

SOLUTION

For $(1 + x)^{1/x}$ to be defined, we must have the following restrictions, $(1 + x) > 0$; to avoid fractional powers of negative numbers, and $x \neq 0$; as we cannot divide by zero. Hence, the natural domain is $(1, \infty) \setminus \{0\}$.

Exercise 0.4.12 The function $f(x) = x^2$ from real numbers to real nonnegative numbers is onto but not one to one.

a. Can you make it one to one by changing its domain? By changing its codomain?

b. Can you make it not onto by changing its domain? Its codomain?

SOLUTION

a. Yes.

   i. Change the domain to $[0, \infty)$. Then $f(x) = x^2$ is one to one and remains onto $\mathbb{R}_{\geq 0}$.

   ii. Changing the codomain to $\{0\}$, makes $f(x) = x^2$ trivially one to one

b. Yes.

   i. Change the domain to $(0, \infty)$. Then 0 is no longer attained, so f is not onto $\mathbb{R}_{\geq 0}$.

   ii. Change the codomain to $\mathbb{R}$. Then negative real numbers are not attained, so f is not onto.

0.5 Real Numbers

Exercise 0.5.1 Show that if p is a polynomial of odd degree with real coefficients, then thereis a real number c such that $p(c) = 0$.

SOLUTION
We will begin by finding two values to contruct the requirements for Intermediate Value Theorem. Firstly, let $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$. Without loss of generality, suppose $a_n > 0$, then $\lim_{x\to-\infty} p(x) = -\infty$ and $\lim_{x\to\infty} p(x) = \infty$; since n is odd. Hence, we can find $N > 0$ large enough such that $p(-N) < 0$ and $p(N) > 0$. By Intermediate Value Theorem, we can find $c \in (-N, N)$ such that $p(c) = 0$.

Exercise 0.5.2 a. Show that the function

$$f(x) = \begin{cases} \sin\frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases} \text{ is not continuous.}$$

b. Show that f satisfies the conclusion of the intermediate value theorem: If $f(x_1) = a_1$ and $f(x_2) = a_2$, then for any number a between a_1 and a_2, there exists a number x between x_1 and x_2 such that $f(x) = a$.

SOLUTION

a. Intuitively, the discontinuity must occur at $x = 0$ since $1/x$ has a discontinuity at the same point. To prove this, we must find a subsequences to show that the limit does not agree at zero. Consider

$$x_n = \frac{1}{\frac{\pi}{2} + 2\pi n} \to 0 \implies f(x_n) = \sin\left(\frac{1}{x_n}\right) = \sin\left(\frac{\pi}{2} + 2\pi n\right) = 1 \neq f(0)$$

Since $\lim_{x\to 0} f(x) \neq f(0)$, f is not continuous.

b. Since f is continuous on $\mathbb{R} \setminus \{0\}$, if both $x_1, x_2 < 0$ or $x_1, x_2 > 0$, then Intermediate Value Theorem applies as it will be continuous on the interval $[x_1, x_2]$ or $[x_2, x_1]$. Hence we just have to prove this for the cases where $x_1 x_2 < 0$.

Without loss of generality, suppose $x_1 < 0 < x_2$. For $\epsilon > 0$, we can find $N > 0$ big enough such that

$$\frac{(4N+3)\pi}{2} > \frac{(4N+1)\pi}{2} > 1/\epsilon$$

Let $\epsilon < \min\{|x_1|, |x_2|\}$. Then we can find an interval $\left[\frac{2}{(4N+3)\pi}, \frac{2}{(4N+1)\pi}\right] \subseteq [x_1, x_2]$. Since

$$f\left(\frac{2}{(4N+1)\pi}\right) = -1, \quad f\left(\frac{2}{(4N+3)\pi}\right) = 1$$

Intermediate Value Theorem applies, thus we can find $f(x) = a$ for $a \in [a_1, a_2] \subseteq [-1, 1]$; since the range of $\sin(1/x)$ is $[-1, 1]$.

Exercise 0.5.3 Suppose $a \leq b$. Show that if $f\ :\ [a, b] \to [a, b]$ is continuous, there exists $c \in [a, b]$ with $f(c) = c$.

SOLUTION
We consider the function $g(x) = f(x) - x$. Then

$$f(x) \in [a, b] \implies a \leq f(x) \leq b$$

so

$$f(a) \geq a \implies g(a) = f(a) - a \geq 0$$

and

$$f(b) \leq b \implies g(b) = f(b) - b \leq 0$$

Since $g(a)g(b) \leq 0$ and g is continuous on $[a, b]$, Intermediate Value Theorem states we can find a $c \in [a, b]$ such that $g(c) = 0$, which means $f(c) = c$.

Exercise 0.5.4 Let

$$a_n = \frac{(-1)^{n+1}}{n}, \quad \text{for } n = 1, 2, \ldots$$

a. Show that the series $\sum a_n$ is convergent.

b. Show that $\sum_{n=1}^{\infty} a_n = \ln 2$.

c. Explain how to rearrange the terms of the series so it converges to 5.

d. Explain how to rearrange the terms of the series so that it diverges.

SOLUTION

a. We will begin by grouping the terms so that we can apply the Monotone Convergence Theorem **(Theorem 0.5.7)**.

$$s_{2n+1} = \sum_{i=1}^{2n+1} a_i = 1 + \sum_{i=1}^{n} a_{2i+1} - \sum_{i=1}^{n} a_{2i} = 1 + \sum_{i=1}^{n} (a_{2i+1} - a_{2i}) = 1 + \sum_{i=1}^{n} \left(\frac{1}{2i+1} - \frac{1}{2i} \right)$$

Since $\frac{1}{2i+1} < \frac{1}{2i}$, then $\frac{1}{2i+1} - \frac{1}{2i} < 0$. So s_{2n+1} is strictly decreasing, and $s_{2n+1} < 1$ for all $n \in \mathbb{N}$. Similarly,

$$s_{2n} = \sum_{i=1}^{2n} a_i = \sum_{i=1}^{n} a_{2i-1} - \sum_{i=1}^{n} a_{2i} = \sum_{i=1}^{n} (a_{2i-1} - a_{2i}) = \sum_{i=1}^{n} \left(\frac{1}{2i-1} - \frac{1}{2i} \right)$$

Since $\frac{1}{2i-1} > \frac{1}{2i}$, then $\frac{1}{2i-1} - \frac{1}{2i} > 0$. So s_{2n} is strictly increasing, and $s_{2n} > 0$ for all $n \in \mathbb{N}$.

Using $0 < s_{2n} < s_{2n+1} < 1$ and by the Monotone Convergence Theorem, (s_{2n+1}) is monotone decreasing and bounded below, thus converging. Similarly, (s_{2n}) is monotone increasing and bounded above, thus converging.

We have shown that the subsequences of odd terms and even terms converge, now we have to show that they converge to the same value. Suppose $(s_{2n}) \to M$ and $(s_{2n+1}) \to L$, then taking limits on both sides of

$$\begin{aligned} |s_{2n+1} - s_{2n}| &= \left| \frac{1}{2n+1} \right| \\ \implies |M - L| &= |\lim a_{2n+1} - s_{2n}| \\ &= \lim \left| \frac{1}{2n+1} \right| = 0 \end{aligned}$$

showing that both subsequence converge to a single value.

b. Consider the Maclaurin Expansion of $\ln(1 + x)$

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots \qquad \text{for } |x| \le 1$$

Substituting $x = 1$, we get

$$\ln 2 = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \sum_{n=1}^{\infty} a_n$$

c. We can construct a series that converges to 5 following this algorithm. If the sum is under 5, the next term in the series will be the next unused odd index, otherwise, the next term will be the next unused even index.

d. We can construct a similar series to above that diverges. Begin by adding odd index terms until the sum is greater than 2. Add the first even index term. Continue adding odd index terms until the sum is greater than 3. Add the second even index term. Repeat by adding odd index terms until the sum is greater than n, then adding the $(n-1)$-th even index term.

Exercise 0.5.5 Show that if a series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent, then any rearrangement of the series is still convergent and converges to the same limit. *Hint:* For any $\epsilon > 0$, there exists N such that $\sum_{n=N+1}^{\infty} < \epsilon$. For any rearrangement $\sum_{k=1}^{\infty} b_n$ of the series, there exists M such that all of $a_1, \dots, a_N$ appear among $b_1, \dots, b_M$. Show that $\left|\sum_{n=1}^{N} a_n - \sum_{n=1}^{M} b_n\right| < \epsilon$.

SOLUTION
Let (b_n) be a rearrangement of (a_n) and $\sum_{n=1}^{\infty} = s$ converges absolutely, meaning $\sum_{n=1}^{\infty} a_n$ converges. We will first show that (b_n) converges absolutely.

For $\epsilon > 0$ we can choose $N > 0$ large enough such that

$$s - \sum_{n=1}^{N} |a_n| < \epsilon \implies \sum_{n=N+1}^{\infty} |a_n| < \epsilon$$

Since N is finite, we can find $M > 0$ such that every term from $a_1, a_2, \dots, a_N$ appears in $b_1, b_2, \dots, b_M$. Formally, we will write $a_k \in \{b_n : n \in \mathbb{N}, n \leq M\}$ for all $k \leq N$. Hence,

$$\begin{aligned}\sum_{n=1}^{M} b_n &= \sum_{a_k \in \{b_1, \dots, b_M\}, k \leq N} a_k + \sum_{a_k \in \{b_1, \dots, b_M\}, k > N} a_k \\ &= \sum_{n=1}^{N} a_n + \sum_{a_k \in \{b_1, \dots, b_M\}, k > N} a_n\end{aligned}$$

Therefore

$$\left|\sum_{n=1}^{M} b_n - \sum_{n=1}^{N} a_n\right| = \left|\sum_{a_k \in \{b_1, \dots, b_M\}, k > N} a_k\right| \leq \sum_{a_k \in \{b_1, \dots, b_M\}, k > N} |a_k| \leq \sum_{n=N+1}^{\infty} |a_n| < \epsilon$$

Let

$$s_N = \sum_{n=1}^{N} a_n, \quad t_M = \sum_{n=1}^{M} b_n$$

Since $s_N \to s$, we get $|s - s_N| < \epsilon$, giving us

$$|t_M - s| \leq |t_M - s_N| + |s_N - s| < \epsilon + \epsilon = \epsilon_0$$

Thus for $M > 0$ large enough, we get

$$\lim_{m \to \infty} t_m = s$$

0.6 Infinite Sets

Exercise 0.6.1 a. Show that the set of rational numbers is countable (i.e., that you can list all rational numbers).

b. Show that the set of $\mathbb{D}$ of finite decimals is countable.

SOLUTION

a. Let

$$A = \{(p, q) \in \mathbb{Z} \times \mathbb{N} : q > 0,\ \gcd(p, q) = 1\}.$$

Every rational number has a unique representation

$$r = \frac{p}{q}, \qquad (p, q) \in A.$$

Since $\mathbb{Z}$ and $\mathbb{N}$ are countable, the Cartesian product $\mathbb{Z} \times \mathbb{N}$ is countable. Hence A, being a subset of $\mathbb{Z} \times \mathbb{N}$, is also countable.

Define

$$f : A \to \mathbb{Q}, \qquad f(p, q) = \frac{p}{q}.$$

By uniqueness of the reduced form, f is one-to-one, and by definition every rational number is of the form p/q with $(p, q) \in A$, so f is onto. Thus f is a bijection, and therefore $\mathbb{Q}$ is countable.

b. Every finite decimal can be written in the form

$$\frac{k}{10^n},$$

where $k \in \mathbb{Z}$ and $n \in \mathbb{N} \cup \{0\}$. Define

$$g : \mathbb{D} \to \mathbb{Z} \times (\mathbb{N} \cup \{0\}), \qquad g\left(\frac{k}{10^n}\right) = (k, n),$$

where the decimal is written without trailing zeros.

This map is one-to-one. Since $\mathbb{Z} \times (\mathbb{N} \cup \{0\})$ is countable, its subset $\mathbb{D}$ is also countable.

Exercise 0.6.2 a. Show that if E is finite and has n elements, then the power set $\mathcal{P}(E)$ has 2^n elements.

b. Choose a map $f : \{a, b, c\} \to \mathcal{P}(\{a, b, c\})$, and compute for that map the set $\{x | x \notin f(x)\}$. Verify that this set is not in the image of f.

SOLUTION

a. Number the elements of E as $e_1, \ldots, e_n$. To form a subset $X \subseteq E$, for each element e_i we have exactly two choices:

$$e_i \in X \qquad \text{or} \qquad e_i \notin X.$$

These choices are independent, so by the multiplication principle there are

$$\underbrace{2 \times 2 \times \cdots \times 2}_{n \text{ times}} = 2^n$$

possible choices. Moreover, each sequence of choices determines a unique subset of E, and every subset arises in this way. Hence $\mathcal{P}(E)$ has exactly 2^n elements.

b. Define $f(x) = \{x\}$. Then $\{x | x \notin f(x)\} = \emptyset$, which is not in the image of f.

Exercise 0.6.3 Show that $(-1,1)$ has the same infinity of points as the reals. *Hint*: Consider the function $g(x) = \tan(\pi x/2)$.

SOLUTION
Considering the function $g(x) = \tan(\pi x/2)$ over the domain $(-1,1)$, we get the mapping $g : (-1,1) \to \mathbb{R}$. We will now show that g is a bijection.

Calculating the derivative of g, gives us $g'(x) = \frac{\pi}{2}\sec^2(\pi x/2) > 0$ for all $x \in (-1,1)$. Since g is continuous and strictly increasing on $(-1,1)$, g is one to one.

To show that g is onto, let

$$\tan\left(\frac{\pi x}{2}\right) = y \implies \frac{\pi x}{2} = \tan^{-1}(y) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

Then

$$-\frac{\pi}{2} < \frac{\pi x}{2} < \frac{\pi}{2} \implies -1 < x < 1 \implies x \in (-1,1)$$

Thus showing that for every $y \in \mathbb{R}$, there exists $x \in (-1,1)$ such that $g(x) = y$, implying that g is onto and therefore a bijection.

We have thus shown that $(-1,1) \asymp \mathbb{R}$.

Exercise 0.6.4 a. How many polynomials of degree d are there with coefficients with absolute value $\leq d$?

b. Give an upper bound for the position in the list of formula 0.6.5 of the polynomial $x^4 - x^3 + 5$.

c. Give an upper bound for the position of the real cube root of 2 is the list of numbers made from the list of formula 0.6.5.

SOLUTION

a. Let $p_d(x) = a_d x^d + a_{d-1}x^{d-1} + \cdots + a_0$ be a polynomial of degree d, with $|a_i| \leq d$ and $a_d \neq 0$.

   To find the number of polynomials, we will consider the number of cases for each coefficient.

   For a_d,

   $$a_d \in \{-d, -d+1, \ldots, d-1, d\} \setminus \{0\}$$

   which can take $(2d)$ possible values.

   For a_i, $i < d$,

   $$a_i \in \{-d, -d+1, \ldots, d-1, d\}$$

   which can take $(2d+1)$ possible values.

   Since each coefficient is independent, by the multiplication principle, there are

   $$(2d) \times \underbrace{(2d+1) \times \cdots \times (2d+1)}_{(d)\text{ times}} = (2d)(2d+1)^d$$

   possible polynomials of degree d.

b. Let P_d be the number of polynomials of degree d or lower, with coefficients with absolute value $\leq d$ and p_i be the polynomial with index position i. Then using similar reasoning to above, $P_d = (2d+1)^{d+1}$.

   While $x^4 - x^3 + 5$ is a polynomial of degree 4, the coefficient of x^0 is 5, hence it must be within the first P_5 polynomials, thus the upper bound for the position of the polynomial $x^4 - x^3 + 5$ is $P_5 = 11^6$.

c. Similarly, $x^3 - 2$ will be within the first P_3 polynomials. Since each polynomial would have at most 3 roots, we will have an upper bound of

   $$3P_3 = 3 \times 7^4$$

   for the position in the list of numbers made from the list of formula 0.6.5.

Exercise 0.6.5 Let $f : A \to B$ and $g : B \to A$ be one to one. We will sketch how to construct a mapping $h : A \to B$ that is one to one and onto.

Let an (f, g)-chain be a sequence consisting alternately of elements of A and B, with elements following an element $a \in A$ being $f(a) \in B$, and the element following an element $b \in B$ being $g(b) \in A$.

a. Show that every (f, g)-chain can be uniquely continued forever to the right, and to the left can

   1. either be uniquely continued forever, or
   2. can be continued to an element of A that is not in the image of g, or
   3. can be continued to an element of B that is not in the image of f.

b. Show that every element of A an every element of B is an element of a unique such maximal (f, g)-chain.

c. Construct $h : A \to B$ by setting

$$h(a) = \begin{cases} f(a) & \text{if } a \text{ belongs to a maximal chain of type 1 or 2 above,} \\ g^{-1}(a) & \text{if } a \text{ belongs to a maximal chain of type 3.} \end{cases}$$

   Show that $h : A \to B$ is well defined, one to one, and onto.

d. Take $A = [-1, 1]$ and $B = (-1, 1)$. It is surprisingly difficult to write a mapping $h : A \to B$. Take $f : A \to B$ defined by $f(x) = x/2$, and $g : B \to A$ defined by $g(x) = x$.

   What elements of $[-1, 1]$ belong to chains of type 1, 2, 3?

   What map $h : [-1, 1] \to (-1, 1)$ does the construction in part c give?

Exercise 0.6.6 Show that the points of the circle $\left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 \mid x^2 + y^2 = 1 \right\}$ have the same infinity of elements as $\mathbb{R}$. *Hint*: This is easy if you use Bernstein's theorem (Exercise 0.6.5)

Exercise 0.6.7 a. Show that $[0, 1) \times [0, 1)$ has the same cardinality as $[0, 1)$.

b. Show that $\mathbb{R}^2$ has the same infinity as $\mathbb{R}$.

c. Show that $\mathbb{R}^n$ has the same infinity as $\mathbb{R}$.

Exercise 0.6.8 Show that the power set $\mathcal{P}(\mathbb{N})$ has the same cardinality as $\mathbb{R}$.

Exercise 0.6.9 Is it possible to list (all) the rationals in $[0, 1]$, written as decimals, so that the entries on the diagonal also give a rational number?